

Assignment : CSS Box Model

- 1] What is the CSS Box Model? Explain its four main components with a diagram.
- The CSS Box model is a fundamental concept in CSS that describes how every HTML element is represented as a rectangular box on a web page.

Each element consists of four main parts:-

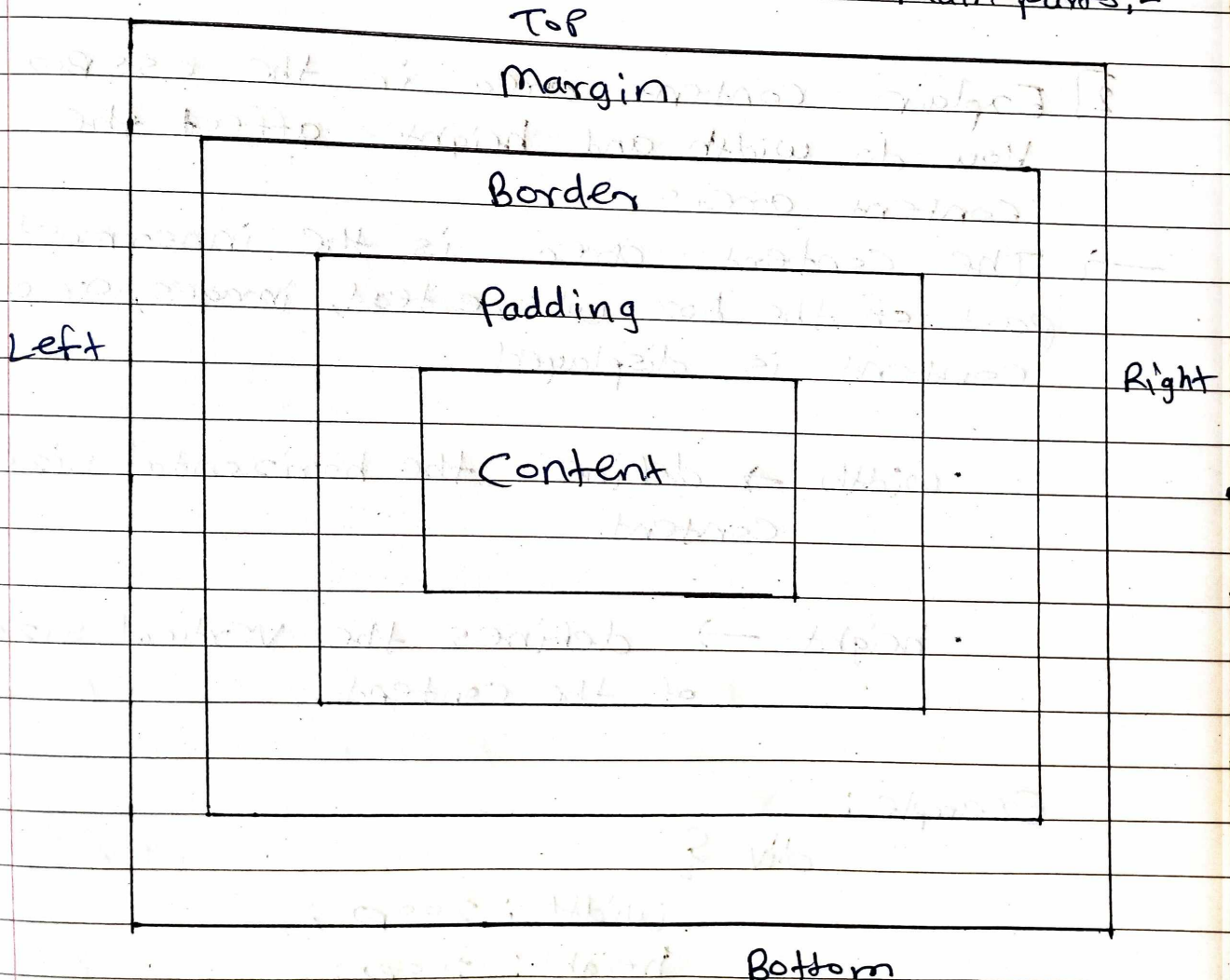


Fig. Box model.

Four components :

1. Content : Actual text, image, or data.
2. padding : Space between Content and border
3. Border : Surrounds padding and content
4. margin : Space outside the border

2] Explain Content area in the CSS Box model.
How do width and height affect the Content area?

→ The Content area is the innermost part of the box where text, images, or other content is displayed.

- width → defines the horizontal size of Content.

- height → defines the vertical size of the content.

Example :

```
div {
```

```
    width : 200px;
```

```
    height : 100px;
```

```
}
```

Padding, border, and margin are added outside the content area (unless box-sizing: border-box is used).

3] What is padding? Write different padding properties and their use.

→ padding is the space between the content and the border.

Padding properties :-

padding : 20px; /* All sides */

padding : 10px 20px; /* Top-Bottom / Left-Right */

padding : 10px 15px 20px 25px;

/* Top / Right / Bottom / Left */

padding-top

padding-right

padding-left

padding-bottom

Use :-

① Improves readability

② prevents content from touching borders

③ Creates internal spacing.

4] Explain border property. write different border styles with examples

→ The border surrounds padding and content.

Border Syntax:

border : width style color;

Border Styles :

border : 2px solid black;

border : 2px dashed red;

border : 2px dotted blue;

border : 4px double green;

Borders are commonly used for;

- cards
- Buttons
- Input fields

5] What is margin? Explain how margin is different from padding.

→ margin is the space outside the border, used to separate elements.

Difference between margin and padding;

Margin	Padding
① Space outside element	① Space inside element
② Affects distance between elements	② Affects content spacing
③ Can collapse	③ Does not collapse

Example :-

margin : 20px;

padding : 20px;

8] What is box-sizing? Explain the difference between content-box and border-box.

→ box-sizing defines how the total width and height of an element are calculated.

Content-box (default) :-

box-sizing : content-box;

• width = content only

• padding + border added outside.

border-box :-

box-sizing : border-box;

- width includes content + padding + border
- Easier layout control.

7] Why is box-sizing : border-box preferred in real projects?

→ Reasons :-

- ① predictable sizing
- ② Easier responsive layouts
- ③ prevents overflow issues.
- ④ commonly used with CSS Frameworks

Example:

```
* {
```

```
  box-sizing : border-box;
```

```
}
```

8] What is margin collapse? when does margin collapse occur?

→ Margin collapse occurs when vertical margins of two elements overlap instead of adding.

Occurs when :

- Two vertical margins meet
- parent and first/last child margins touch.
- only vertical margins collapse (not horizontal).

Example : —

<p> First </p>

<p> Second </p>

If both have margin : 20px,
actual gap = 20px, not 40px.

3) Explain margin shorthand with examples.
Margin shorthand Formats : —

margin : 20px; /* All sides */

margin : 10px 20px; /* Top-Bottom /
Left / Right */

margin : 10px 15px 20px 25px;
/* Top / Right / Bottom / Left */

Shorthand reduces code and improves readability.

- Box-model defines Spacing and layout
- Padding is internal space
- margin is external space
- border-box simplifies layouts.

Q) Write real-life use cases of CSS Box model in web page layouts.

Real-life uses :

- ① Creating Cards and Containers
- ② Spacing between sections.
- ③ Button design
- ④ Responsive layouts
- ⑤ Forms and input spacing
- ⑥ UI consistency across pages.

Almost every website layout ~~is~~ relies on the CSS Box model.